LESSON PLAN

NDT (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Noise suppression and its levels
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Noise suppression and its levels. Various methods of thrust reversal and principles of operation.
5. Specific Objectives	20.1.2.1 Trainee will be able to understand and appreciate noise separation methods, thrust reversal mechanism, after burning fuel system, lubrication system, thrust augmentation devices, cooling and thrust calculations

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of noise suppression diagrams and thrust reversal principles from IB/PDF; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary source of noise in a jet engine?
(b) Define noise suppression in the context of aviation.
(c) List two methods used for thrust reversal.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Noise suppression methods • Noise levels and standards • Thrust reversal mechanisms • Principles of thrust reversal operation	L & Ds	PPT, IB	15 min
(b)	• Afterburning fuel system • Lubrication system overview • Thrust augmentation devices • Cooling and thrust calculations	L & Ds	PPT, PDF	20 min
(c)	• Noise separation techniques • Thrust reversal safety limits • System integration review	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how thrust reversal aids in landing safety.
(b) Describe the role of noise suppression in modern aircraft design.

12. Home Assignment and Follow Up	Prepare a brief note on the relationship between thrust augmentation devices and noise generation for the next lesson on lubrication systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Lubrication system main components and its purposes, operation of each components of system, Air cooling and sealing of engine main bearings	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min